MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU – UGP (HONOURS) REGULAR EXAMINATION, MARCH 2026
Fourth Semester

MG4DSCSTA204– STATISTICS

STATISTICAL MODELLING IN DATA SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain] [CO No(s)]

SCHEME OF VALUATION

Part A

Short Answer Questions

Answer any **seven** questions. Each question carries **2 marks**.

1. Definition of data or mention any two types of data or any two examples – 2 marks [K][1]
2. Any two examples – 2 marks (Sales, banking, E- commerce, logistics, healthcare, human resources etc) or definition of transactional data only – 1 mark. [U][1]
3. One example each for discrete data and continuous data or definition of both – 2 marks [U][1]
4. Definition of both data or giving one example for each data – 2 marks [U][1]
5. Mentioning any one example or any one use of time series data or defining time series data – 2 marks [U][1]
6. Definition of test criterion / test statistic – 2 marks. [K][2]
7. Definition of both – 2 marks [U][2]
8. Any two assumptions – 2 marks (Normality, independence, homogeneity of variances,…) [An][3]
9. Definition or any two advantages or any two applications – 2 marks [U][4]
10. Getting Y = 51 - 2 marks [An][4] (7x2=14 marks)

Part B

Short Essay Questions

Answer any **four** questions. Each question carries **6 marks**.

11. Explaining both or giving some examples of both or mentioning some characteristics of both – 6 marks [U][1]
12. Mentioning how it is applied in both cases – 6 marks (Targeting recommendation: Whatever the user searches on internet, he/she will see numerous posts everywhere. Delivery logistics: Data science helps the companies to find the best route for the shipment of the products, the best time suited for delivery, the best mode of transport to reach the destination etc) [U][1]
13. Mentioning how it is applied in both cases – 6 marks. (Environmental science: Spatial data can help identify critically endangered habitats and track changes in land use, allowing for targeted conservation efforts. Understanding spatial distribution of variables like temperature and precipitation is crucial for modeling climate change etc. Technology: Navigation apps like Google map, augmented reality) [U][1]

14. Stating the hypotheses (at least null hypothesis) H0: $\mu = 1500$ against H1: $\mu \neq 1500$ - 2 marks

Test statistic under H0, $Z = \frac{(\underline{X} - \mu_0)}{\sigma}\sqrt{n} = \frac{(1494 - 1500)}{2}\sqrt{90} = -28.4605$

- 2 marks

Conclusion: Critical region $|Z| > 1.96$ and hence reject H0 - 2 marks [A][2]

15. Explaining goodness of fit testing – 6 marks or give 2 marks to each sub question [U][3]
16. Advantages – 3 marks; limitations – 3 marks. Explaining linear regression only – 6 marks[U][4]

(4x6=24 marks)

Part C

Essay Type Questions

Answer any **one** question. Each question carries **12 marks**.

17. Explaining any three types of biological data – 4 + 4 + 4 marks (sequence data, structure data, expression data, interaction data, phenotype data) or generally explaining biological data with some examples – 12 marks [U][1]

18. Calculation of the test statistic under H0, $\chi^2 = \sum_i \left\{\frac{(O_i - E_i)^2}{E_i}\right\} = 40.9699$

- 5 marks

where expected frequencies are 64.8, 70.2, 55.2, 59.8 – 1 mark each

OR

Calculation of the test statistic under H0, $\chi^2 = \sum_i \left\{\frac{(O_i - E_i)^2}{E_i}\right\} = 40.32426$

- 5 marks

where expected frequencies are 65, 70, 55, 60 – 1 mark each

Degrees of freedom = (r-1) (c-1) = 1 - 1 mark

Conclusion: Critical region $\chi^2 > 3.841$ and hence reject H0 - 2 marks [A][3] (1x12=12 marks)